Project Progress Report: Round-Trip Vision-Language Navigation

Platform: Unitree Go2 quadruped navigation in Isaac Sim

1. Baseline Findings: The Original Problem Assumption Was Incomplete

The project initially assumed that Return failures were mainly caused by long-horizon localisation drift and confusion between the Outbound, Confirm, and Return stages. However, the clean NaVILA baseline showed that the more immediate limitation is the stability of NaVILA's own closed-loop navigation.

Clean NaVILA baseline	**Result**
Outbound success	14/30
Return success	5/30
Complete round-trip success	3/30

This baseline used explicit phase switching and dataset-provided reverse instructions, so it directly reflects NaVILA's closed-loop navigation ability in continuous Isaac Sim environments. Among episodes that completed Outbound, only about 35% completed Return. Typical failures included missing the stopping point, oscillating between left and right actions, making little effective progress, and stopping at the wrong location. Therefore, the dominant practical problem cannot be explained only by Return-stage drift or subtask-boundary confusion.

2. External Route Memory and Oracle Feasibility Validation

During Outbound, the external memory stores sparse anchors at approximately 1 m intervals along the travelled route. Each anchor contains route order, geometry relative to neighbouring anchors, and a local LiDAR map. During Return, the system estimates the robot's position relative to the current and next anchors, converts the next anchor's distance and bearing into a compact route hint, and updates anchor progress until the robot reaches the start.

The Oracle version keeps this pipeline unchanged, but replaces relocalisation and route-progress estimation with the robot's absolute pose and anchor world coordinates from Isaac Sim. Visual understanding, VLM action selection, local navigation, and low-level control remain autonomous; Oracle does not provide a full action sequence or continuously drive the robot.

With Oracle guidance, the Return success rate reached 17/18 (94.4%). This demonstrates that the route-memory-hint-VLM framework can support end-to-end round-trip navigation when the route reference is reliable.

The Oracle system also includes a conservative HintActionArbiter, which intervenes only when the VLM action clearly conflicts with the hint and the hinted direction is confirmed to be obstacle-free.

Return-phase decision	**Count**
VLM action consistent with hint	180
No correction because hinted direction was obstructed	153
Temporary corrective intervention	15
Total	348

Only 4.3% of decisions were corrected, showing that the system did not rely on Oracle for continuous control. A small number of targeted interventions improved Return reliability while preserving substantial VLM autonomy.

3. Progress of the Non-Oracle Route Relocalisation System

A non-Oracle relocalisation module is being developed to replace Oracle route progress and hints. It uses the LiDAR anchors recorded during Outbound and matches the live point cloud only against the current and next anchors using ICP. The module currently runs in parallel with Oracle: the VLM still receives Oracle hints, while the non-Oracle output is evaluated independently for accuracy and stability.

Recent non-Oracle evaluation	**Result**
Current-anchor exact accuracy	65.3%
Next-anchor exact accuracy	65.5%
Overshoot	18.6%
Next-behind: robot physically passed next anchor	0.05%
Median distance error	0.075 m
Median bearing error	5.64 degrees

Anchor progression is now largely stable, and genuine cases of missing the next anchor are extremely rare. The main remaining problem is ICP ambiguity in repetitive corridors, similar walls, and rotationally symmetric structures. These scenes can produce multiple plausible matches, and some incorrect solutions still receive high overlap, inlier count, and confidence scores.

4. Remaining Work and Plan

Remaining work	**Plan**
Detect high-confidence incorrect matches	Add ambiguity-aware ICP scoring, near-tie detection, and neighbouring-anchor consistency checks
Reduce single-frame instability	Aggregate geometric evidence across consecutive frames or local submaps
Replace Oracle hints	Connect non-Oracle outputs to the VLM and run closed-loop Return evaluation
Final validation	Compare directly with Oracle, complete batch testing, failure analysis, dissertation results, and demonstration video